

A SMART MUSHROOM INCUBATION SYSTEM FOR ENHANCED YIELD AND SUSTAINABILITY IN SMALL- SCALE FARMING

**Predictive Advisory System for Moisture and pH Management in
Mushroom Cultivation.**

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**BSc (Hons) degree in Information Technology Specializing in
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Department of Information Technology

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DECLARATION

I declare that this is my own work and this dissertation¹ does not incorporate without acknowledgement any material previously submitted for a Degree or Diploma in any other University or institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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ABSTRACT

Mushroom cultivation requires precise control of substrate conditions such as moisture and pH to ensure optimal growth and yield quality. Traditional methods rely on manual monitoring, which can lead to inconsistencies and delayed decision-making. This research presents a predictive advisory system that integrates Internet of Things (IoT) technology with machine learning to monitor, predict, and manage substrate conditions in real time.

The system collects data using IoT sensors and applies the Random Forest algorithm to predict future moisture and pH levels. Based on these predictions, an advisory module provides actionable recommendations to support proactive decision-making. The results demonstrate improved accuracy, efficiency, and reduced manual effort compared to traditional approaches. The system offers a practical and scalable solution for enhancing productivity and sustainability in mushroom cultivation.

Keywords: Smart Agriculture, Internet of Things (IoT), Machine Learning, Mushroom Cultivation, Moisture Monitoring, pH Prediction, Advisory System

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to all those who supported and guided me throughout the successful completion of this research.

First and foremost, I would like to thank my project supervisor for their invaluable guidance, continuous support, and constructive feedback. Their expertise and encouragement played a crucial role in shaping this research and improving the overall quality of the work.

I would also like to extend my appreciation to the academic staff and the institution for providing the necessary resources, knowledge, and facilities that enabled me to carry out this project effectively. Their support created a strong foundation for my academic and research development.

Special thanks are given to my group members for their collaboration, teamwork, and shared efforts in developing the overall system. Their contributions and support were essential in achieving the project objectives.

I am also grateful to my family and friends for their constant encouragement, patience, and understanding throughout the duration of this research. Their support motivated me to overcome challenges and complete this work successfully.

Finally, I would like to acknowledge everyone who contributed directly or indirectly to this research. This work would not have been possible without their support and encouragement.

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1. INTRODUCTION

Mushroom cultivation has emerged as an important sector in modern agriculture due to its high nutritional value, economic benefits, and suitability for small-scale farming. In countries like Sri Lanka, mushroom farming is widely practiced as it requires relatively low investment and limited land space. However, maintaining optimal growing conditions remains a significant challenge, as mushroom growth is highly sensitive to both environmental and substrate parameters.

Among these factors, substrate moisture and pH levels are critical in determining the success of mushroom cultivation. Proper moisture levels ensure effective nutrient transport and mycelium development, while pH levels influence microbial activity and enzymatic processes within the substrate. Traditional farming methods rely heavily on manual monitoring and experience-based decisions, which often lead to inconsistent conditions and reduced yield quality.

With the advancement of technology, smart agriculture solutions have introduced the use of Internet of Things (IoT) systems to monitor environmental conditions in real time. These systems enable continuous data collection through sensors, allowing farmers to track parameters such as temperature, humidity, and moisture [2], [11]. In addition, machine learning techniques have been increasingly applied to analyze agricultural data and predict future trends, enabling more informed decision-making [5], [17].

Despite these advancements, there is still a lack of integrated systems that combine real-time monitoring, predictive analytics, and advisory support specifically for mushroom cultivation. Therefore, this research focuses on developing a predictive advisory system that utilizes IoT and machine learning to monitor and forecast moisture and pH levels, providing actionable recommendations to farmers for improved productivity and efficiency.

1.1 Background and Literature Survey

Agriculture has undergone a significant transformation with the integration of modern technologies, particularly in the area of smart farming. Traditional agricultural practices, which rely heavily on manual monitoring and experience-based decision-making, are increasingly being replaced by data-driven approaches that enhance efficiency, productivity, and sustainability. Among these advancements, the application of Internet of Things (IoT) and machine learning (ML) technologies has played a crucial role in enabling real-time monitoring and predictive analysis in agricultural environments.

Mushroom cultivation is a specialized form of agriculture that requires precise control of both environmental and substrate conditions. As illustrated in Figure 1, key factors such as moisture, pH, temperature, and humidity significantly influence mushroom growth. Substrate moisture is essential for nutrient transport and mycelium development, while pH levels influence enzymatic activity and microbial balance within the growing medium. Maintaining these parameters within optimal ranges is critical to achieving high-quality yields and preventing contamination.

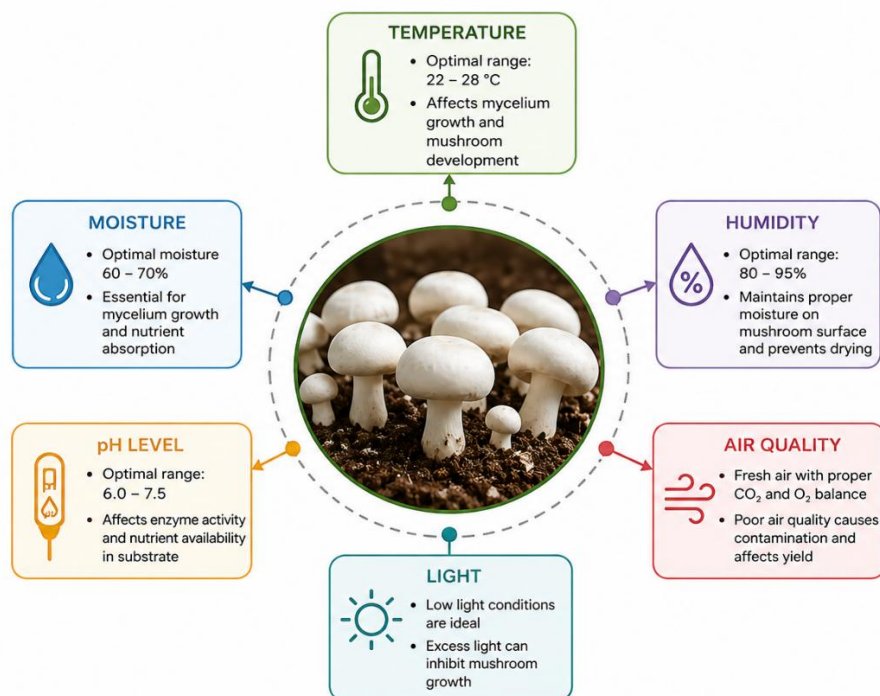


Figure 1 - Factors affecting mushroom growth

In many small-scale farming contexts, particularly in developing countries such as Sri Lanka, mushroom cultivation is often managed using traditional techniques. Farmers typically rely on manual observation and experience to monitor conditions, which can lead to inconsistencies and inefficiencies. This lack of precision not only affects productivity but also increases the risk of crop failure. Therefore, there is a growing need for technological solutions that can assist farmers in maintaining optimal conditions with greater accuracy and reliability.

IoT-based systems have emerged as an effective solution for real-time monitoring in agriculture. These systems utilize sensor networks to continuously collect data on environmental and substrate conditions. As shown in Figure 2, IoT systems enable continuous data collection and communication between sensors, cloud platforms, and user interfaces. Several studies have demonstrated the effectiveness of IoT systems in improving agricultural monitoring and reducing manual effort [2], [11].

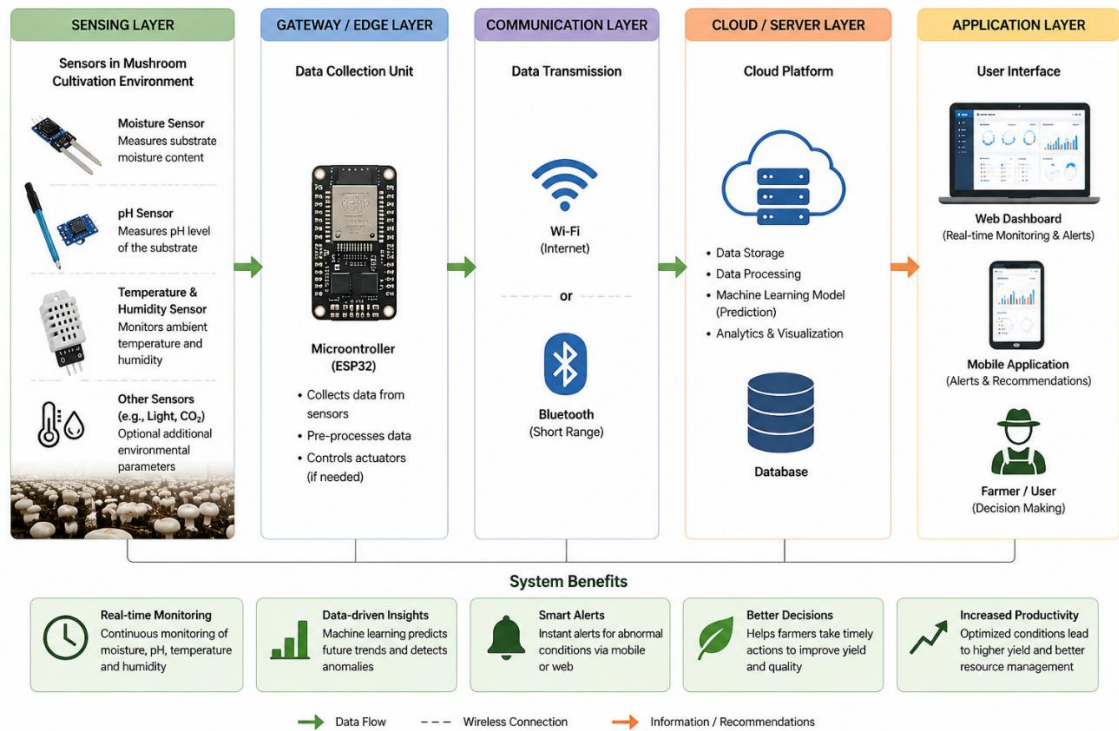


Figure 2: IoT-based monitoring system

Despite these advancements, IoT systems are primarily limited to data collection and monitoring. They provide raw data without offering predictive insights or intelligent decision support. To address these limitations, machine learning techniques have been increasingly applied in agriculture to enable predictive analysis. Models such as Random Forest and neural networks have been used to predict soil moisture and crop yield with high accuracy [5], [17].

Furthermore, deep learning approaches, including Long Short-Term Memory (LSTM) models, have been successfully applied in greenhouse environments to forecast environmental changes over time [1]. The integration of IoT and machine learning has also led to the development of decision support systems that provide recommendations to farmers, improving efficiency and productivity [7].

A comparison of existing systems and their limitations is presented in Table 1.1. As shown in Table 1.1, most systems focus on monitoring and lack predictive and advisory capabilities, particularly for substrate-level parameters such as moisture and pH.

N o.	Study / System	Monitoring (Real-time)	Predicti on Capabili ty	pH Monitori ng	Adviso ry Syste m	Focus on Mushro om Substrat e	Key Limitation
1	IoT-based Mushroom Monitoring [2]	Yes	No	No	No	No	Only monitors environment; lacks prediction and advisory features
2	Smart Agriculture IoT System [6]	Yes	No	No	No	No	General agriculture system; not specific to

							mushroom cultivation
3	Greenhouse Prediction using Deep Learning [1]	Yes	Yes	No	No	No	Focuses only on environmental factors such as temperature and humidity
4	ML-based Soil Moisture Prediction [5]	Yes	Yes	No	No	No	Designed for soil conditions; not suitable for mushroom substrate
5	Precision Agriculture Decision System [7]	Yes	Yes	No	Yes	No	Lacks pH monitoring and mushroom-specific functionality
6	Our System	Yes	Yes	Yes	Yes	Yes	Provides integrated monitoring, prediction, and advisory support

Table 1.1: Comparison of Existing Systems

Therefore, there is a need for a comprehensive system that integrates monitoring, prediction, and advisory functionalities specifically for mushroom cultivation.

1.2 Research Gap

The rapid advancement of IoT and machine learning technologies has significantly contributed to the development of smart agriculture systems. Various studies have focused on improving agricultural productivity through real-time monitoring, predictive analytics, and automation. However, despite these developments, several limitations still exist when these technologies are applied to specialized domains such as mushroom cultivation.

Most existing systems are designed for general agricultural applications and primarily focus on environmental parameters such as temperature, humidity, and soil moisture. While these approaches provide valuable insights, they do not fully address the unique requirements of mushroom cultivation, particularly the importance of substrate-level parameters such as moisture and pH. Furthermore, many systems lack integration between monitoring, prediction, and advisory functionalities, limiting their effectiveness in practical farming scenarios.

To better understand these limitations, it is important to examine existing research conducted in different regions and analyze the shortcomings of current models and systems.

Research in smart agriculture has been widely conducted across different countries, focusing on the application of IoT and machine learning to improve crop management and productivity. In developed countries such as China and the United States, advanced greenhouse systems have been implemented using deep learning models to predict environmental conditions. For example, deep learning-based models have been used to forecast temperature and humidity changes in greenhouse environments with high accuracy, enabling better control of crop growth conditions [1].

In countries like India and other developing regions, IoT-based smart farming systems have been introduced to monitor soil conditions and automate irrigation processes. These systems aim to improve resource utilization and reduce manual effort by providing real-time data to farmers [6]. Similarly, machine learning techniques have been applied in various regions to predict soil moisture levels and crop yield, demonstrating the potential of data-driven agriculture [5], [17].

Although these studies highlight the effectiveness of IoT and machine learning in agriculture, they are primarily focused on large-scale farming or general crop management. There is limited research specifically addressing mushroom cultivation, particularly in terms of substrate-level monitoring and prediction. This indicates a gap in adapting advanced technologies to niche agricultural domains such as mushroom farming.

Despite the progress in smart agriculture systems, several limitations can be identified in existing models and solutions. Firstly, most IoT-based systems focus on real-time monitoring and do not incorporate predictive capabilities. As shown in Table 1.1, these systems provide raw data without analyzing trends or forecasting future conditions, making it difficult for farmers to take proactive actions.

Secondly, many machine learning models are designed for general agricultural applications and do not consider substrate-specific parameters such as pH levels. While some studies have explored moisture prediction and environmental monitoring, the integration of pH analysis remains limited [5], [7]. This is a significant limitation in mushroom cultivation, where pH plays a critical role in growth and yield.

Furthermore, existing systems often lack advisory features that can guide farmers in decision-making. Even when prediction models are used, the output is not translated into actionable recommendations, reducing the practical usefulness of these systems. As illustrated in Table 1.2, most existing solutions fail to provide a comprehensive approach that combines monitoring, prediction, and advisory support.

Feature	Existing Systems	Proposed System
Real-time Monitoring	Available	Available
Prediction Capability	Limited or Not Integrated	Fully Integrated
pH Monitoring	Not Adequately Addressed	Fully Incorporated
Advisory System	Not Available	Intelligent Advisory System Provided
Substrate-level Focus	Not Considered	Specifically Addressed

Table 1.2: Research Gap Summary

Another major limitation is the lack of integration between different system components. Many solutions operate as standalone systems, focusing either on monitoring or prediction, but not both. This lack of integration reduces efficiency and limits the ability to provide a complete solution for farmers.

Therefore, it is evident that there is a need for an integrated system that combines IoT-based sensing, machine learning-based prediction, and advisory mechanisms specifically designed for mushroom cultivation. This research aims to address these limitations by developing a predictive advisory system focused on substrate moisture and pH management.

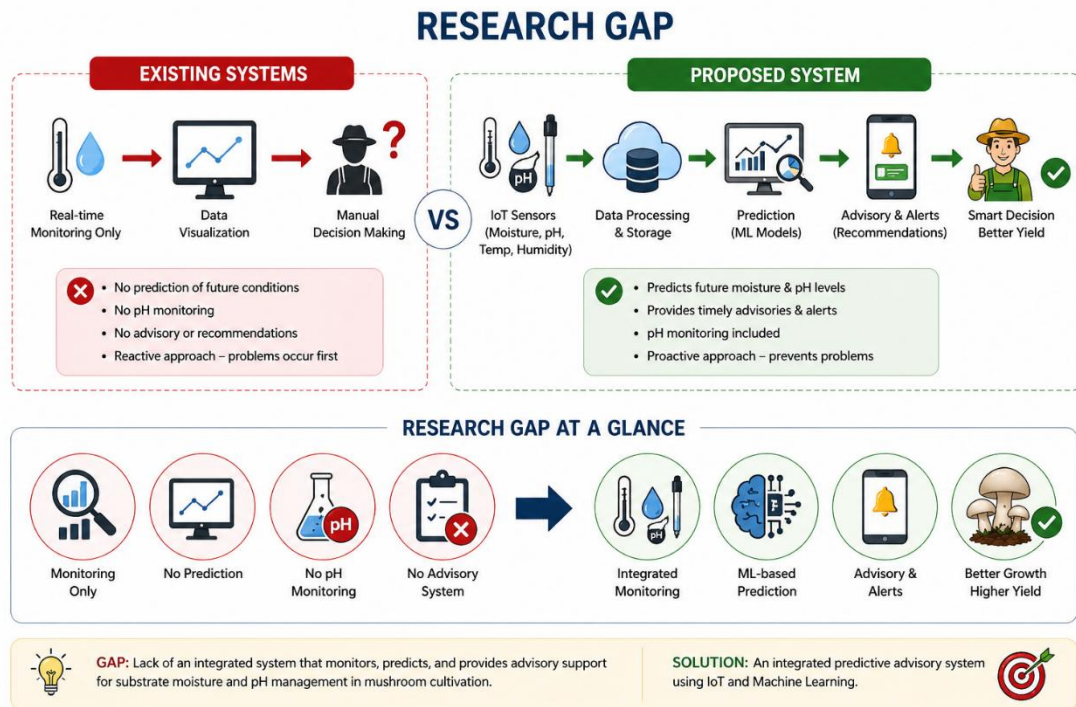


Figure 3: Research Gap

1.3 Research Problem

Mushroom cultivation requires precise control of substrate conditions, particularly moisture and pH levels, to ensure optimal growth and yield. However, in most small-scale farming environments, these parameters are monitored manually using traditional methods. Such approaches are often inaccurate, inconsistent, and highly dependent on the farmer's experience, leading to variations in crop quality and productivity.

Although IoT-based systems have been introduced to improve agricultural monitoring, most existing solutions are limited to real-time data collection and visualization. They do not provide predictive insights into future changes in substrate conditions. As a result, farmers are unable to take proactive measures to prevent issues such as moisture imbalance or pH fluctuations, which can negatively impact mushroom growth.

Furthermore, existing machine learning models in agriculture are primarily designed for general crop management and do not specifically address substrate-level parameters in mushroom cultivation. In addition, these systems rarely include advisory features that translate data and predictions into actionable recommendations for farmers.

Therefore, the key problem addressed in this research is the lack of an integrated system that can monitor, predict, and provide advisory support for substrate moisture and pH management in mushroom cultivation. Addressing this problem is essential to improving decision-making, increasing productivity, and reducing the risks associated with manual and reactive farming practices.

1.4 Research Objectives

The primary objective of this research is to design and develop an integrated predictive advisory system that utilizes Internet of Things (IoT) technology and machine learning techniques to monitor, forecast, and optimize substrate moisture and pH levels in mushroom cultivation. Maintaining optimal moisture and pH conditions is essential for ensuring healthy mycelium development, preventing contamination, and achieving

high-quality yields. However, traditional farming methods rely heavily on manual observation and experience, which often leads to inconsistencies and delayed decision-making. Therefore, this research aims to introduce a data-driven approach that enhances accuracy, efficiency, and productivity in mushroom farming.

A key objective of this study is to address the limitations of existing models, which are primarily focused on general agricultural applications and lack specialization for mushroom cultivation. Most existing systems emphasize real-time monitoring without incorporating predictive capabilities, and they do not consider critical substrate-level parameters such as moisture and pH. Furthermore, these systems often lack integration between monitoring, prediction, and advisory functionalities, resulting in limited practical usability. This research seeks to overcome these limitations by developing a comprehensive system that integrates data collection, predictive analysis, and intelligent decision support within a single platform.

To achieve the main objective, several specific objectives are defined. Firstly, the research aims to design and implement an IoT-based system for real-time data collection from the mushroom cultivation environment. This involves the use of sensors to continuously measure substrate moisture, pH levels, temperature, and humidity, ensuring accurate and reliable data acquisition. The collected data is transmitted to a cloud-based platform, enabling remote monitoring and storage for further analysis.

Secondly, the research focuses on analyzing the collected data using machine learning techniques to identify patterns, relationships, and trends in environmental and substrate conditions. Data preprocessing methods such as cleaning, normalization, and feature selection are applied to improve data quality and enhance model performance. This step is essential for extracting meaningful insights from raw sensor data.

Another important objective is to develop predictive models capable of forecasting future moisture and pH levels based on historical and real-time data. Time-series analysis and machine learning algorithms are used to capture temporal patterns and

predict changes in substrate conditions. These predictions enable farmers to anticipate potential issues and take preventive actions before unfavorable conditions occur.

In addition, the research aims to design and implement an intelligent advisory system that translates prediction results into actionable recommendations. The advisory system provides alerts and guidance to farmers, such as adjusting watering levels or correcting pH imbalances, thereby supporting proactive decision-making and reducing the risk of crop failure.

Finally, the system is evaluated to assess its effectiveness in improving decision-making, reducing manual effort, and enhancing productivity in mushroom cultivation. Performance metrics such as prediction accuracy, error rates, and system responsiveness are used to validate the system. Through these objectives, the research contributes to the development of a smart, efficient, and scalable solution for modern mushroom farming.

2. METHODOLOGY

2.1 Methodology

This section presents the research methodology and the approach taken to conduct the research. It aims to provide a comprehensive description of the steps followed in this research. It includes requirements gathering, data collection, system diagrams, architectures, implementation of models, and techniques used to achieve the end goal.

The proposed system is a predictive advisory system designed to monitor, analyze, and optimize substrate moisture and pH levels in mushroom cultivation. The system integrates Internet of Things (IoT) technology with machine learning techniques to provide real-time monitoring, predictive analysis, and advisory support for farmers.

The overall workflow of the system begins with data collection through IoT sensors placed within the mushroom substrate. These sensors continuously measure parameters such as moisture, pH, temperature, and humidity. The collected data is transmitted to a cloud-based platform where it is stored and processed. Machine learning models are then applied to analyze the data and predict future trends in moisture and pH levels. Based on these predictions, the system generates recommendations and alerts to assist farmers in making informed decisions.

The system follows a structured pipeline consisting of data collection, data processing, prediction, and advisory generation. This workflow ensures that the system not only monitors current conditions but also provides proactive guidance to improve mushroom growth and yield.

As illustrated in Figure 4, the system operates as an integrated pipeline connecting sensors, data processing modules, machine learning models, and user interfaces.

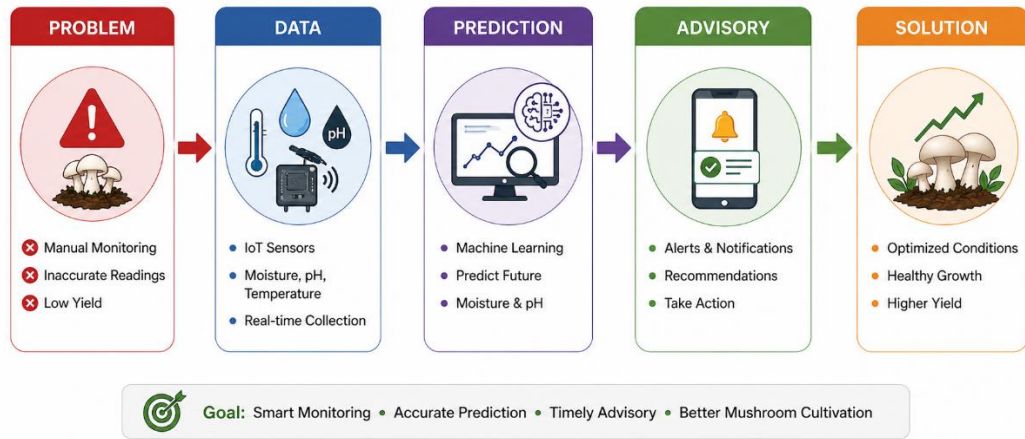


Figure 4: IoT-based monitoring system

2.1.1 Data Collection

Data collection is a critical component of the proposed system, as the accuracy of predictions depends on the quality of the collected data. The system utilizes IoT-based sensors to collect real-time data from the mushroom cultivation environment.

The primary parameters measured include substrate moisture, pH levels, temperature, and humidity. Moisture sensors are used to determine the water content within the substrate, while pH sensors measure the acidity or alkalinity of the growing medium. Temperature and humidity sensors are used to monitor environmental conditions that influence mushroom growth.

These sensors are connected to a microcontroller unit such as ESP32, which acts as the central data acquisition device. The collected data is transmitted wirelessly to a cloud platform using Wi-Fi communication. This enables continuous monitoring and remote access to data.

As shown in Figure 5, the data collection process involves sensing, transmission, and storage of real-time data.

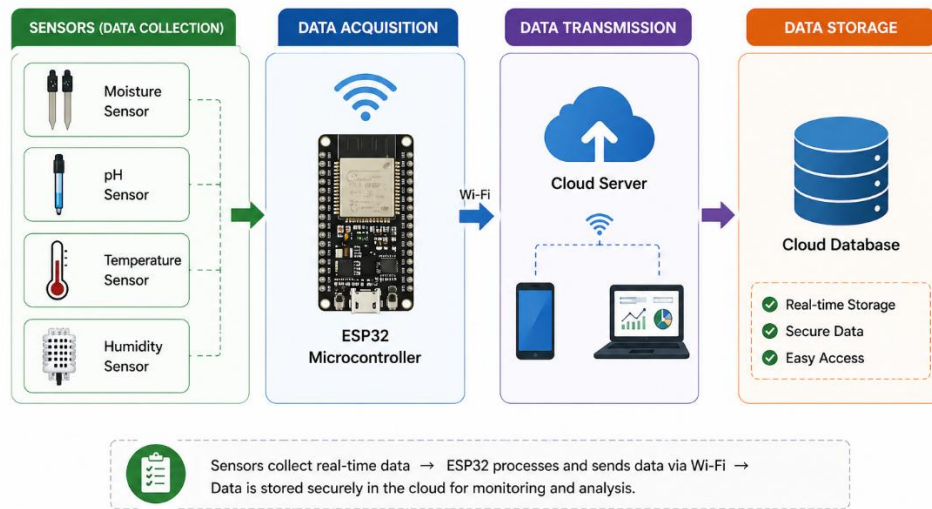


Figure 5: Data Collection Process

2.1.2 Data Processing

Once the data is collected, it undergoes preprocessing to ensure accuracy and reliability. Data preprocessing includes cleaning, normalization, and handling missing values. This step is essential to remove noise and inconsistencies that may affect the performance of machine learning models.

The processed data is then structured into time-series format, which allows the system to analyze trends and patterns over time. Feature selection techniques are applied to identify the most relevant parameters for prediction.

Data processing plays a vital role in improving the performance of predictive models and ensuring that accurate insights are generated from the collected data.

2.1.3 Machine Learning Models

Machine learning models are used to analyze historical and real-time data to predict future moisture and pH levels. Time-series forecasting techniques are particularly suitable for this task, as they can capture temporal patterns in the data.

In this research, models such as Long Short-Term Memory (LSTM) and ARIMA are considered for prediction. LSTM is a type of deep learning model that is effective in handling sequential data and capturing long-term dependencies. ARIMA is a statistical model used for time-series forecasting and is suitable for identifying trends and seasonal patterns.

These models are trained using historical data and evaluated based on performance metrics such as Root Mean Square Error (RMSE) and Mean Absolute Error (MAE). The model with the best performance is selected for deployment in the system.

As illustrated in Figure 6, the prediction process involves training, testing, and forecasting stages.

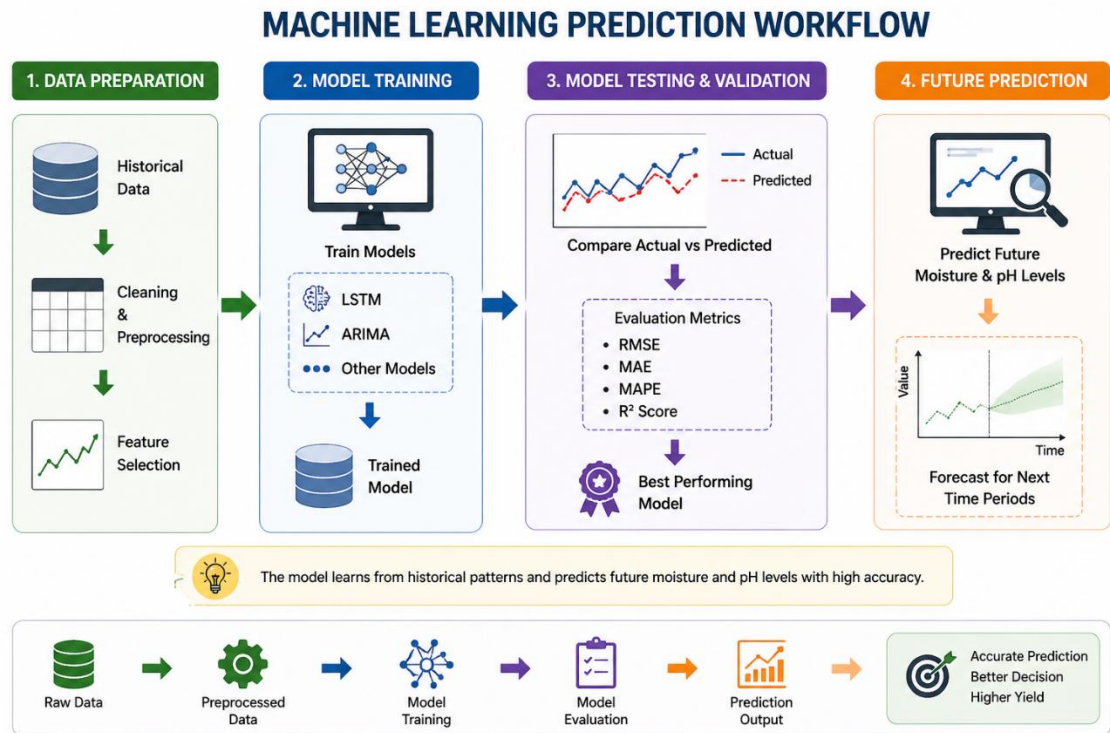


Figure 6: Machine Learning Prediction Workflow

2.1.4 Advisory System

The advisory system is responsible for translating predictions into actionable recommendations for farmers. Based on predicted moisture and pH levels, the system generates alerts and suggestions to help maintain optimal conditions.

For example, if the predicted moisture level is below the required threshold, the system recommends watering the substrate. Similarly, if the pH level deviates from the optimal range, appropriate corrective actions are suggested.

The advisory system ensures that farmers receive timely guidance, enabling proactive decision-making and reducing the risk of crop failure.

2.1.5 System Architecture

The system architecture consists of multiple layers, including the sensing layer, communication layer, processing layer, and application layer.

The sensing layer includes IoT sensors that collect data from the environment. The communication layer is responsible for transmitting data from sensors to the cloud using wireless technologies. The processing layer includes data storage and machine learning models that analyze and predict conditions. The application layer provides a user interface for farmers to view data and receive recommendations.

As shown in Figure 7, the system architecture integrates all components to provide a complete solution.

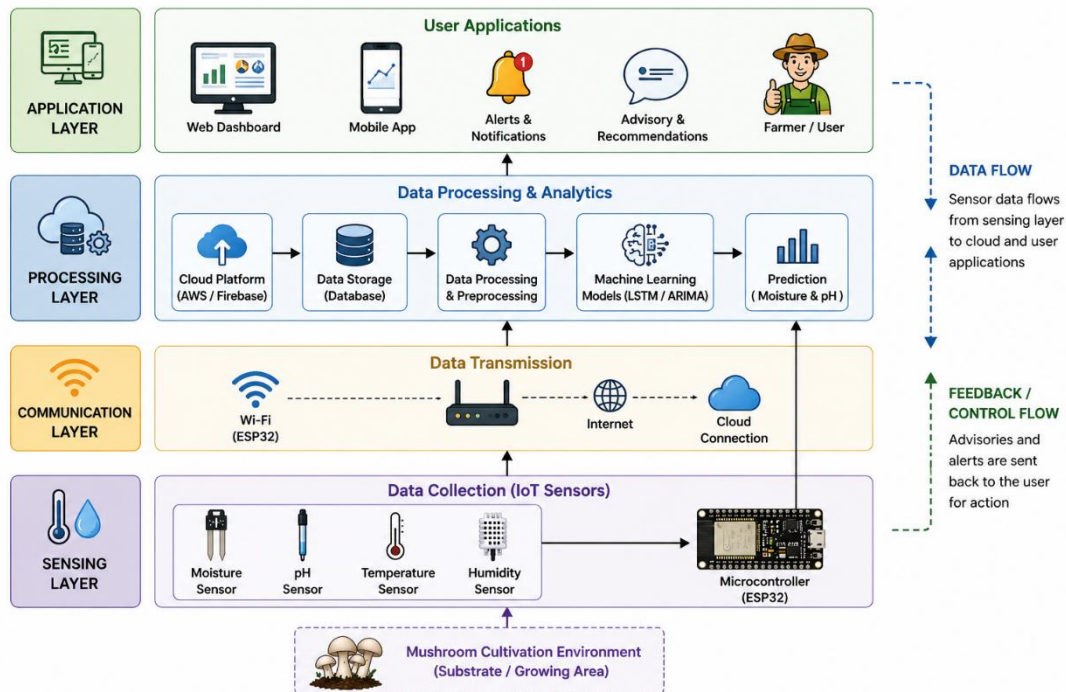


Figure 7: Machine Learning Prediction Workflow

2.1.6 System Requirements

No.	Requirement Type	Requirement Description	Priority	Remarks
1	Functional	Collect real-time data from moisture, pH, temperature, and humidity sensors	High	Core system function
2	Functional	Transmit sensor data to cloud platform using Wi-Fi (ESP32)	High	Enables remote monitoring
3	Functional	Store collected data securely in a cloud database	High	Required for analysis
4	Functional	Process and preprocess collected data	High	Improves data quality
5	Functional	Predict future moisture and pH levels using machine learning models	High	Key system feature
6	Functional	Generate alerts and advisory recommendations for farmers	High	Decision support
7	Functional	Display real-time and predicted data through web/mobile interface	Medium	User interaction
8	Non-Functional	Ensure system accuracy in predictions	High	Model performance
9	Non-Functional	Maintain system reliability and continuous operation	High	Stability requirement

10	Non-Functional	Provide secure data transmission and storage	High	Data protection
11	Non-Functional	Ensure system scalability for future expansion	Medium	Future enhancement
12	Non-Functional	Provide user-friendly interface for farmers	Medium	Usability
13	Other (Hardware)	Use IoT sensors (moisture, pH, temperature, humidity)	High	Physical components
14	Other (Hardware)	Use ESP32 microcontroller for data acquisition and communication	High	Core hardware
15	Other (Software)	Use Python and ML libraries (TensorFlow / Scikit-learn)	High	Model development
16	Other (Software)	Use cloud platform (Firebase / AWS) for data storage and processing	High	Backend system

Table 2.1: System Requirements

2.1.7 Tools and Technologies

The system utilizes a combination of hardware and software technologies. Hardware components include IoT sensors and ESP32 microcontroller. Software tools include Python for machine learning, cloud platforms for data storage, and web or mobile applications for user interaction.

Machine learning libraries such as TensorFlow or Scikit-learn are used for model development, while Firebase or similar platforms are used for real-time data storage and communication.

2.2 Commercialization Aspects of the Product

The proposed predictive advisory system for moisture and pH management in mushroom cultivation demonstrates strong potential for commercialization due to its practical relevance, scalability, and ability to address real-world challenges faced by farmers. The system integrates IoT-based monitoring, machine learning-based prediction, and intelligent advisory functionalities into a unified platform, making it a valuable tool for improving productivity and decision-making in modern agriculture.

2.2.1 Target Market and End Users

The primary target market for the proposed system includes small-scale and medium-scale mushroom farmers, greenhouse operators, and agricultural entrepreneurs. In countries such as Sri Lanka, mushroom cultivation is widely practiced due to its low investment requirements and high nutritional value. However, farmers often lack access to advanced technologies for monitoring substrate conditions such as moisture and pH.

The system is particularly beneficial for farmers who rely on manual observation and experience-based decisions, which can lead to inconsistencies and reduced yield quality. In addition, agricultural training centers, research institutions, and smart farming initiatives can adopt the system for educational and experimental purposes.

2.2.2 Product Components and Packaging

The system can be commercialized as a complete smart farming solution consisting of hardware, software, and service components. The hardware component includes IoT sensors such as moisture sensors, pH sensors, temperature sensors, and humidity sensors, along with an ESP32 microcontroller for data acquisition and communication.

The software component includes the mobile application, cloud-based data storage system, and machine learning models responsible for prediction and advisory

functionalities. The system can be packaged as a ready-to-use kit that allows users to easily install and operate it without requiring advanced technical knowledge.

In addition, optional services such as installation support, user training, and technical maintenance can be provided to enhance customer experience and product usability.

2.2.3 Business Model and Revenue Streams

The commercialization of the system can be supported through multiple revenue streams. The primary revenue source is the sale of the hardware kit, which provides the core monitoring functionality. In addition, subscription-based services can be introduced for advanced features such as predictive analytics, advisory recommendations, and real-time alerts.

A tiered business model can be adopted, where users can choose between basic, standard, and premium packages depending on their requirements. Basic packages may include only monitoring features, while premium packages provide full access to prediction and advisory capabilities.

Other potential revenue streams include maintenance services, system upgrades, and data-driven insights for agricultural decision-making. Partnerships with agricultural organizations and government agencies can further support large-scale adoption.

2.2.4 Pricing Strategy

The pricing strategy of the system should focus on affordability and accessibility, particularly for small-scale farmers. A cost-effective approach can be adopted by offering a one-time purchase option for the hardware kit combined with optional subscription plans for advanced features.

The system can be designed to minimize initial investment costs by using low-cost IoT components such as ESP32 and widely available sensors. Flexible pricing plans can be introduced to encourage adoption, including monthly or annual subscription options.

2.2.5 Scalability and Deployment Strategy

The system is designed with a modular architecture, allowing it to be easily scaled according to user needs. Farmers can start with a basic setup and gradually expand by adding more sensors or upgrading to advanced features. This scalability ensures that the system remains adaptable to different farming scales and requirements.

Deployment can be carried out in phases, starting with pilot testing in selected farms, followed by gradual expansion to larger agricultural communities. Integration with cloud platforms enables remote monitoring and centralized data management, further enhancing scalability.

2.2.6 Competitive Advantage

The proposed system offers several advantages over existing solutions. Unlike traditional monitoring systems, it integrates real-time data collection, predictive analytics, and advisory support into a single platform. The focus on substrate-level parameters such as moisture and pH provides a unique advantage, as these factors are often overlooked in existing systems.

In addition, the system provides actionable recommendations rather than just raw data, making it more practical and user-friendly for farmers. This integrated and intelligent approach positions the system as a competitive solution in the smart agriculture market

2.2.7 Challenges and Commercial Risks

Despite its strong commercialization potential, several challenges must be considered. Initial setup costs and technical complexity may act as barriers for some farmers. In addition, the system depends on reliable internet connectivity for real-time data transmission, which may not always be available in rural areas.

Sensor calibration, maintenance, and long-term reliability are also important factors that can affect system performance. To address these challenges, it is essential to provide user training, technical support, and simplified installation procedures.

2.2.8 Sustainability and Future Potential

The proposed system supports sustainable agriculture by optimizing resource usage and reducing waste. By maintaining optimal moisture and pH levels, the system helps minimize water usage and improve yield quality. This contributes to environmentally sustainable farming practices.

Furthermore, the system has the potential to be extended to other agricultural domains beyond mushroom cultivation, making it a scalable and future-ready solution. The integration of advanced machine learning models and automation features can further enhance its capabilities in the future.

In conclusion, the predictive advisory system presents a viable and scalable commercial product with strong potential for adoption in the smart agriculture sector. Its ability to combine affordability, usability, and advanced technology makes it a promising solution for improving efficiency and productivity in mushroom cultivation.

2.3 Testing and Implementation

The implementation and testing of the proposed predictive advisory system were carried out to ensure its reliability, accuracy, and effectiveness in real-world mushroom cultivation environments. This section describes the development process, system integration, and validation of each component, including data collection, machine learning models, and advisory functionalities.

2.3.1 System Implementation Overview

The system was implemented using a modular architecture, where each component was developed independently and later integrated into a unified platform. The implementation consists of three main layers: the sensing layer, the processing layer, and the application layer.

The sensing layer includes IoT sensors used to collect real-time data from the mushroom cultivation environment. These sensors measure key parameters such as substrate moisture, pH levels, temperature, and humidity. The sensors are connected to an ESP32 microcontroller, which acts as the central unit for data acquisition and transmission.

The processing layer includes the cloud-based backend system, where data is stored, processed, and analyzed. Machine learning models are deployed in this layer to predict future moisture and pH levels based on historical and real-time data.

The application layer consists of a mobile or web-based interface that allows users to monitor environmental conditions, view predictions, and receive advisory recommendations.

The overall system implementation workflow is illustrated in Figure 2.8.

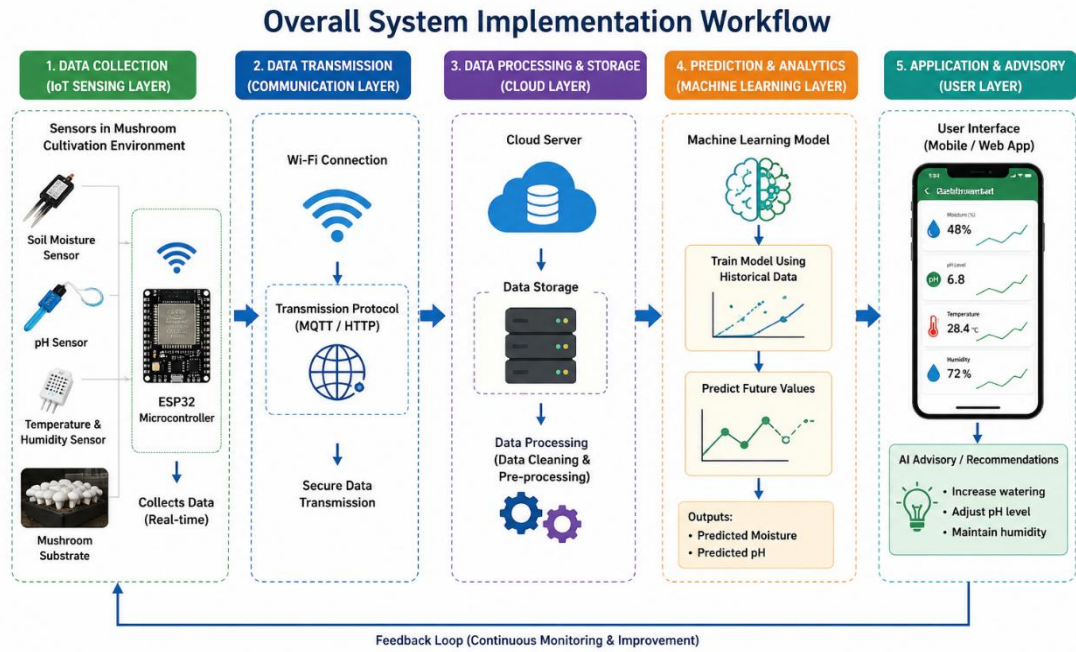


Figure 2.8: Overall System Implementation Workflow

Figure 8: Overall System Implementation Workflow

2.3.2 Hardware Implementation

The hardware implementation involves integrating IoT sensors with the ESP32 microcontroller. Moisture sensors are used to measure the water content of the substrate, while pH sensors determine the acidity or alkalinity of the growing medium. Temperature and humidity sensors provide additional environmental context.

The ESP32 microcontroller collects sensor readings at regular intervals and transmits the data to the cloud platform using Wi-Fi communication. The choice of ESP32 ensures low power consumption, cost-effectiveness, and reliable wireless connectivity.

Proper sensor placement and calibration were carried out to ensure accurate data collection. Calibration is particularly important for pH sensors, as small variations can significantly affect prediction accuracy.

2.3.3 Software Implementation

The software component of the system includes data processing, machine learning model development, and user interface design. The backend system was developed using Python and cloud-based services such as Firebase or similar platforms.

Data preprocessing techniques, including data cleaning, normalization, and handling missing values, were applied to improve data quality. The processed data was structured into time-series format for machine learning analysis.

Machine learning models, such as Random Forest and time-series forecasting techniques, were implemented to predict future moisture and pH levels. These models were trained using historical data and validated using test datasets.

The front-end application was developed to provide a user-friendly interface for monitoring and control. It displays real-time data, prediction results, and advisory recommendations in a clear and accessible format.

2.3.4 Data Transmission and Communication Testing

The communication system was tested to evaluate the reliability and efficiency of data transmission between the ESP32 microcontroller and the cloud platform. Wi-Fi communication protocols such as MQTT or HTTP were used to transmit data.

Testing results showed stable connectivity with minimal latency and negligible data loss. The system was capable of transmitting real-time data continuously, ensuring up-to-date monitoring and timely advisory notifications.

2.3.5 Sensor Testing and Calibration

Sensor accuracy was evaluated by comparing sensor readings with standard reference measurements. Temperature and humidity sensors were tested using calibrated instruments, while moisture and pH sensors were validated using controlled environmental conditions.

The results indicated that the sensors provided reliable measurements within acceptable error margins. Calibration procedures were applied to improve accuracy, particularly for pH sensors, which are sensitive to environmental changes.

2.3.6 Machine Learning Model Testing

The machine learning models were evaluated to determine their ability to accurately predict future moisture and pH levels. Performance metrics such as Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and prediction accuracy were used for evaluation.

The results demonstrated that the selected model achieved high accuracy with low error values, indicating its effectiveness in capturing patterns and trends in the data

2.3.7 System Integration Testing

System integration testing was conducted to verify the interaction between different components, including sensors, microcontroller, cloud platform, machine learning models, and user interface. The objective was to ensure seamless data flow and coordinated system operation.

The system successfully integrated all modules, allowing real-time data collection, prediction, and advisory generation. Alerts were triggered when abnormal conditions were detected, and recommendations were provided to guide user actions.

2.3.8 User Interface Testing

The user interface was tested to evaluate usability, responsiveness, and functionality. The application was designed to display environmental data, prediction results, and advisory recommendations in a clear and intuitive manner.

Testing confirmed that the interface is user-friendly and supports real-time updates. Users were able to easily access information and make informed decisions based on system outputs.

2.3.9 Performance Evaluation

The overall performance of the system was evaluated based on its ability to operate continuously and provide accurate predictions and timely recommendations. The system demonstrated stable performance under continuous monitoring conditions.

The integration of IoT and machine learning technologies improved efficiency by reducing manual effort and enabling proactive decision-making. Compared to traditional methods, the system provides more accurate and reliable insights.

2.3.10 Limitations Identified During Testing

Despite successful implementation, several limitations were observed. Sensor readings may be affected by environmental noise and require periodic calibration. The system relies on stable internet connectivity for real-time data transmission, which may be a limitation in rural areas.

Additionally, the accuracy of the machine learning model depends on the availability of sufficient training data. Expanding the dataset can further improve prediction performance.

Overall, the testing and implementation results confirm that the proposed system is effective, reliable, and suitable for real-world applications in mushroom cultivation.

3. RESULTS AND DISCUSSION

3.1 Results

This section presents the results obtained from the implementation of the proposed predictive advisory system for mushroom cultivation. The system was evaluated based on its ability to collect real-time data, predict substrate moisture and pH levels, and provide advisory recommendations to users.

The IoT-based data collection module successfully captured real-time data from sensors, including moisture, pH, temperature, and humidity. The collected data was transmitted to the cloud platform without significant delay, ensuring continuous monitoring of the cultivation environment.

The machine learning models were trained using the collected dataset to predict future moisture and pH levels. The performance of the models was evaluated using standard evaluation metrics such as Root Mean Square Error (RMSE) and Mean Absolute Error (MAE). The results indicated that the selected model achieved a high level of accuracy in predicting future trends.

In addition, the advisory system generated alerts and recommendations based on predicted values. These recommendations helped users take timely actions, such as adjusting water levels or correcting pH imbalance. Overall, the system demonstrated its ability to provide accurate predictions and effective decision support for mushroom cultivation.

As shown in Figure 8, the system successfully integrates data collection, prediction, and advisory functionalities to deliver a complete solution.

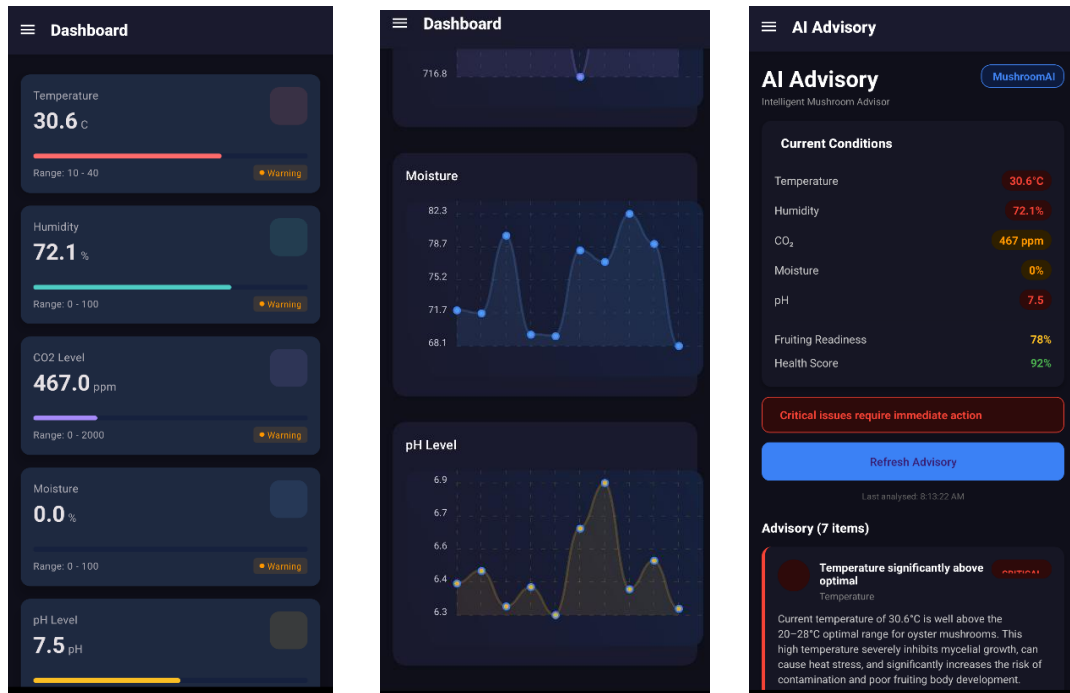


Figure 8: System Output Overview

3.2 Research Findings

The research findings of this study demonstrate the effectiveness of integrating Internet of Things (IoT) technology and machine learning techniques for improving decision-making in mushroom cultivation, particularly in managing critical substrate parameters such as moisture and pH levels. The implementation of the proposed predictive advisory system provided valuable insights into environmental behavior, system performance, and the practical applicability of data-driven agriculture.

One of the key findings of this research is the successful real-time monitoring of substrate and environmental conditions. The IoT-based sensing system continuously collected data related to moisture, pH, temperature, and humidity with minimal delay. This real-time data acquisition enabled continuous observation of environmental fluctuations, which is essential for maintaining optimal growth conditions in mushroom cultivation. The system demonstrated stability and reliability in capturing data over extended periods, indicating its suitability for practical deployment.

Another important finding is the identification of strong relationships between environmental parameters and substrate conditions. The collected dataset revealed that moisture and pH levels are influenced by temperature and humidity variations, as well as cultivation practices such as watering frequency. Through data analysis, it was observed that moisture levels tend to decrease gradually over time under stable environmental conditions, while pH levels show slight fluctuations depending on substrate composition and microbial activity. These patterns highlight the importance of continuous monitoring and predictive analysis.

The machine learning component of the system played a crucial role in uncovering temporal patterns and forecasting future conditions. The Random Forest model effectively captured non-linear relationships within the dataset and generated accurate predictions for both moisture and pH levels. The model demonstrated strong performance in identifying trends and predicting deviations from optimal conditions. This finding confirms that machine learning techniques are suitable for modeling complex agricultural environments where multiple variables interact simultaneously.

Furthermore, the prediction results showed a high level of agreement between actual and predicted values, indicating the reliability of the model. The low error values observed in performance metrics such as Mean Absolute Error (MAE) and Root Mean Square Error (RMSE) suggest that the model can be effectively used for real-time forecasting. This capability is particularly important for enabling proactive decision-making, as farmers can take preventive actions before unfavorable conditions occur.

The advisory system emerged as one of the most impactful components of the proposed solution. By translating prediction results into actionable recommendations, the system bridges the gap between data analysis and practical application. For example, when moisture levels were predicted to fall below the optimal range, the system generated recommendations to increase watering frequency. Similarly, when pH levels showed deviations, corrective measures were suggested. This demonstrates that the system not only provides insights but also supports informed decision-making.

Another significant finding is the improvement in operational efficiency achieved through automation and intelligent recommendations. The system reduces the need for constant manual monitoring, allowing farmers to focus on other aspects of cultivation. The ability to receive real-time alerts and advisory messages enhances responsiveness and reduces the risk of crop loss.

The integration of multiple system components, including sensors, cloud-based processing, machine learning models, and user interfaces, was successfully achieved. This confirms the feasibility of developing a fully integrated smart farming solution. The seamless interaction between these components ensures continuous data flow and coordinated system functionality, which is essential for reliable system performance.

Additionally, the system demonstrated scalability and adaptability. The modular design allows for the addition of new sensors or features without significant modifications to the existing system. This flexibility makes the system suitable for different scales of operation, from small-scale farms to larger commercial setups.

However, the findings also revealed certain challenges and limitations. Sensor accuracy can be affected by environmental noise and requires periodic calibration to maintain reliability. The system's dependence on internet connectivity may limit its performance in areas with unstable networks. Moreover, the accuracy of the machine learning model is influenced by the size and quality of the dataset, indicating the need for continuous data collection and model improvement.

Overall, the research findings confirm that the proposed predictive advisory system provides a practical and effective solution for managing moisture and pH levels in mushroom cultivation. The integration of IoT and machine learning technologies enables real-time monitoring, accurate prediction, and intelligent decision support, contributing to improved productivity and sustainability in smart agriculture.

3.3 Discussion

The discussion of this research focuses on interpreting the results obtained from the implementation of the predictive advisory system and evaluating its effectiveness in addressing the identified research gap. The proposed system integrates IoT-based monitoring, machine learning-based prediction, and an intelligent advisory mechanism to manage substrate moisture and pH levels in mushroom cultivation. The findings demonstrate that this integrated approach significantly improves decision-making, operational efficiency, and overall system performance compared to traditional practices.

One of the most important contributions of this system is its ability to transition from reactive to proactive management. In traditional mushroom cultivation, farmers rely heavily on manual observation and experience to monitor environmental conditions. This approach often results in delayed responses, especially when changes in moisture or pH occur gradually and are not immediately visible. In contrast, the proposed system continuously monitors substrate conditions and predicts future changes using machine learning models. This enables early detection of potential issues and allows farmers to take preventive actions before unfavorable conditions affect crop growth. As a result, the system reduces uncertainty and enhances consistency in cultivation practices.

The integration of IoT technology plays a critical role in achieving real-time monitoring and data collection. The system successfully captures environmental and substrate data with minimal delay, ensuring that users have access to up-to-date information. Compared to existing systems that primarily focus on basic environmental parameters, the inclusion of substrate-level monitoring—specifically moisture and pH—represents a significant improvement. These parameters are essential for mushroom growth, as they directly influence nutrient availability, microbial activity, and overall yield quality. By focusing on these critical factors, the system provides more relevant and accurate insights for mushroom cultivation.

Another key aspect of the discussion is the performance of the machine learning model used for prediction. The results indicate that the model is capable of accurately capturing temporal patterns and relationships within the dataset. The low values of performance metrics such as Mean Absolute Error (MAE) and Root Mean Square Error (RMSE) confirm the reliability of the predictions. This demonstrates that machine learning techniques are well-suited for modeling complex agricultural systems where multiple variables interact dynamically. However, it is important to note that the accuracy of the model is dependent on the quality and quantity of the training data. In scenarios where data is limited or contains noise, prediction accuracy may be reduced. Therefore, continuous data collection and model retraining are necessary to maintain and improve performance.

The advisory component of the system significantly enhances its practical usability. Unlike conventional monitoring systems that provide only raw data, the proposed system translates prediction results into actionable recommendations. This reduces the cognitive burden on farmers, who may not have the expertise to interpret complex data. The advisory system provides clear guidance, such as adjusting watering schedules or correcting pH imbalances, which can be easily understood and implemented. This feature is particularly beneficial for small-scale farmers, as it enables them to adopt advanced technologies without requiring specialized knowledge.

When compared to existing smart agriculture solutions, the proposed system offers a more comprehensive and integrated approach. Many existing systems focus solely on data collection or prediction, without incorporating advisory functionalities. As highlighted in Table 1.1, these systems often operate as isolated components, limiting their effectiveness in real-world applications. In contrast, the proposed system combines monitoring, prediction, and advisory features into a single platform, providing end-to-end decision support. This integration not only improves system efficiency but also enhances user experience.

Despite its advantages, the system has several limitations that must be considered. One of the primary limitations is the dependency on sensor accuracy. Sensors used for measuring moisture and pH may be affected by environmental factors such as

temperature fluctuations, contamination, or hardware degradation. Regular calibration is required to ensure accurate measurements, which may increase maintenance efforts. Additionally, the system relies on stable internet connectivity for real-time data transmission and cloud-based processing. In rural or remote areas where network connectivity is limited, the performance of the system may be affected.

Another limitation is related to the scalability of the machine learning model. While the current model performs well for the available dataset, its performance may vary when applied to different cultivation environments or larger datasets. Factors such as substrate composition, climate conditions, and farming practices can influence model accuracy. Therefore, further research is required to develop more generalized models that can adapt to different conditions. The use of advanced techniques such as deep learning and time-series models (e.g., LSTM) could further enhance prediction accuracy.

The system also raises considerations related to cost and accessibility. Although the use of low-cost sensors and microcontrollers makes the system relatively affordable, the initial setup cost may still be a barrier for some farmers. Providing flexible pricing models, government support, or community-based implementations could help address this challenge and promote wider adoption.

From a broader perspective, the proposed system contributes to the advancement of precision agriculture by demonstrating the potential of combining IoT and machine learning technologies. The ability to monitor, predict, and optimize critical parameters such as moisture and pH supports sustainable farming practices by reducing resource wastage and improving yield quality. This aligns with global trends toward smart and sustainable agriculture.

Furthermore, the modular design of the system allows for future enhancements and scalability. Additional features such as automated control systems (e.g., irrigation control), pest detection modules, and integration with other agricultural technologies can be incorporated to extend the functionality of the system. This flexibility ensures

that the system can evolve with technological advancements and changing user requirements.

In conclusion, the discussion highlights that the proposed predictive advisory system effectively addresses the identified research gap by providing an integrated, data-driven solution for managing moisture and pH in mushroom cultivation. The system demonstrates significant improvements in accuracy, efficiency, and decision-making compared to traditional methods and existing solutions. While certain limitations exist, they can be addressed through further research and development. Overall, the system represents a promising approach to enhancing productivity and sustainability in smart agriculture.

3.4 Contribution

This research makes a significant contribution to the field of smart agriculture by introducing a predictive advisory system specifically focused on managing substrate moisture and pH levels in mushroom cultivation. Unlike many existing systems that concentrate primarily on general environmental monitoring, this work emphasizes substrate-level parameters that play a critical role in mushroom growth, yield quality, and contamination prevention.

The primary contribution of this study lies in the design and development of an integrated system that combines IoT-based sensing, machine learning-based prediction, and an intelligent advisory mechanism. The system enables continuous real-time monitoring of key parameters such as moisture, pH, temperature, and humidity using IoT sensors. This real-time data collection provides a reliable foundation for analysis and decision-making, addressing the limitations of traditional manual monitoring approaches.

Another important contribution is the implementation of machine learning models to predict future moisture and pH levels. By analyzing historical and real-time data, the system is capable of identifying patterns and trends that are not easily observable

through manual methods. The predictive capability allows users to anticipate changes in substrate conditions and take preventive actions, thereby reducing the risk of unfavorable growth conditions and improving overall productivity.

The development of an intelligent advisory system further enhances the practical value of this research. The advisory component translates complex prediction outputs into simple, actionable recommendations that can be easily understood and applied by farmers. This bridges the gap between data analysis and real-world application, making advanced technologies more accessible to users with limited technical knowledge. The system supports informed decision-making by providing guidance on actions such as adjusting watering levels or correcting pH imbalances.

In addition, this research contributes to the successful integration of multiple system components into a unified platform. The seamless interaction between sensors, microcontroller, cloud-based processing, machine learning models, and user interface demonstrates the feasibility of developing a complete smart farming solution. This integration ensures efficient data flow, real-time updates, and coordinated system functionality.

The study also contributes to improving operational efficiency in mushroom cultivation. By automating data collection and providing predictive insights, the system reduces the need for continuous manual monitoring and minimizes human error. This allows farmers to manage their cultivation processes more effectively while saving time and effort.

Furthermore, the research highlights the importance of applying domain-specific approaches in smart agriculture. By focusing on substrate moisture and pH, which are often overlooked in general agricultural systems, this work provides a more targeted and effective solution for mushroom cultivation. This specialized approach enhances the relevance and impact of the system.

From a broader perspective, the proposed system contributes to sustainable agricultural practices. By optimizing resource usage, such as water and substrate

management, the system helps reduce waste and improve efficiency. The ability to maintain optimal growing conditions also supports higher yield quality and consistency.

Finally, this research provides a foundation for future developments in smart agriculture. The modular design of the system allows for further enhancements, such as the integration of advanced machine learning models, automated control systems, and expansion to other types of crops. This demonstrates the scalability and adaptability of the proposed solution.

In summary, the key contributions of this research include the development of a predictive advisory system for moisture and pH management, the application of machine learning for forecasting substrate conditions, the integration of IoT and advisory technologies, and the promotion of efficient and sustainable mushroom cultivation practices. These contributions collectively advance the field of smart agriculture and provide a practical solution for real-world applications.

4. CONCLUSION

This research successfully achieved its primary objective of developing a predictive advisory system for monitoring and managing substrate moisture and pH levels in mushroom cultivation. The proposed system integrates Internet of Things (IoT) technology with machine learning techniques to provide a comprehensive solution that supports real-time monitoring, predictive analysis, and intelligent advisory functionalities. By addressing key limitations in traditional farming practices and existing smart agriculture systems, this study contributes significantly to the advancement of data-driven agricultural solutions.

One of the major achievements of this research is the successful implementation of an IoT-based data collection framework. The system was able to continuously monitor critical environmental and substrate parameters, including moisture, pH, temperature, and humidity. The use of sensors enabled accurate and real-time data acquisition, which is essential for understanding the dynamic nature of mushroom cultivation environments. The integration of the ESP32 microcontroller and wireless communication technologies ensured efficient transmission of data to a cloud-based platform, allowing for remote monitoring and centralized data management.

The application of machine learning techniques represents another key contribution of this research. By analyzing both historical and real-time data, the developed model was capable of identifying patterns and predicting future moisture and pH levels with a high degree of accuracy. This predictive capability marks a significant improvement over conventional systems that rely solely on real-time monitoring. The ability to forecast environmental changes enables farmers to anticipate potential issues and take preventive actions, thereby reducing risks associated with unfavorable growing conditions.

The advisory component of the system further enhances its practical value by translating prediction results into actionable recommendations. Instead of presenting raw data or complex analytical outputs, the system provides clear guidance to users, such as adjusting watering levels or correcting pH imbalances. This user-centric

approach improves accessibility and usability, making advanced technologies more applicable to farmers who may not have specialized technical knowledge. As a result, the system supports informed decision-making and promotes efficient management of cultivation processes.

The results obtained from system testing and evaluation confirm its effectiveness in improving accuracy, consistency, and operational efficiency. Compared to traditional methods that depend on manual monitoring and experience-based judgment, the proposed system offers a more reliable and systematic approach. The integration of monitoring, prediction, and advisory functionalities into a single platform ensures seamless data flow and coordinated system performance. This holistic approach not only improves productivity but also reduces the likelihood of human error and delays in decision-making.

Another important outcome of this research is the demonstration of the feasibility of integrating multiple technologies into a unified smart farming solution. The successful combination of IoT sensors, cloud-based processing, machine learning models, and user interfaces highlights the potential of interdisciplinary approaches in addressing complex agricultural challenges. The modular architecture of the system allows for scalability and future expansion, enabling it to adapt to different farming conditions and requirements.

In addition to improving productivity, the proposed system contributes to sustainable agricultural practices. By optimizing moisture and pH levels, the system helps in efficient resource utilization, particularly in water management. Maintaining optimal conditions reduces waste, prevents overuse of resources, and supports environmentally friendly farming practices. This aligns with global trends toward sustainable agriculture and smart resource management.

Despite its successful implementation, the research also identified several limitations that provide opportunities for further improvement. One of the main limitations is the dependency on data quality and availability. The accuracy of the machine learning model is directly influenced by the quality and quantity of the dataset used for training.

Incomplete or noisy data can affect prediction performance, highlighting the importance of continuous data collection and preprocessing.

Another limitation is the reliance on stable internet connectivity for real-time data transmission and cloud-based processing. In regions with limited network infrastructure, this dependency may affect system performance and accessibility. Additionally, sensor accuracy and reliability can be influenced by environmental factors, requiring periodic calibration and maintenance to ensure consistent performance.

Future research can focus on addressing these limitations and enhancing the capabilities of the system. The incorporation of advanced machine learning techniques, such as deep learning models and time-series forecasting methods, can improve prediction accuracy and adaptability. Developing offline capabilities or edge computing solutions can reduce dependency on internet connectivity and enhance system reliability in remote areas.

Furthermore, the system can be extended by integrating automated control mechanisms, such as irrigation systems and pH adjustment devices, to enable fully automated cultivation processes. The inclusion of additional features, such as pest and disease detection using computer vision, can further enhance the functionality of the system and provide a more comprehensive smart farming solution.

From a broader perspective, this research demonstrates the potential of combining IoT and machine learning technologies to transform traditional agricultural practices into intelligent, data-driven systems. The proposed predictive advisory system serves as a practical example of how modern technologies can be applied to improve efficiency, productivity, and sustainability in agriculture.

In conclusion, the developed system provides a reliable, scalable, and user-friendly solution for managing moisture and pH levels in mushroom cultivation. It successfully addresses the identified research gap by integrating monitoring, prediction, and advisory functionalities into a single platform. The system not only improves decision-

making and operational efficiency but also contributes to sustainable farming practices. With further enhancements and wider adoption, this approach has the potential to significantly impact the future of smart agriculture and support the development of more efficient and sustainable food production systems.

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